

Media Engineering and Technology Faculty
German University in Cairo



Personalized Multimodal Mnemonics in Immersive VR Memory Palaces: Design and Experimental Evaluation of Space, Content, Visual, and Auditory Mnemonics Personalization

Master's Thesis – First Draft

Current study: personalised versus generic virtual memory palaces

Author:	Mark Emeel Sroor
Supervisors:	Dr. Yomna Mahmoud Tobias Wagner (advisory role; confirm designation)
Draft Date:	Working draft – September 2026

Media Engineering and Technology Faculty
German University in Cairo



Personalized Multimodal Mnemonics in Immersive VR Memory Palaces: Design and Experimental Evaluation of Space, Content, Visual, and Auditory Mnemonics Personalization

Master's Thesis – First Draft

Current study: personalised versus generic virtual memory palaces

Author:	Mark Emeel Sroor
Supervisors:	Dr. Yomna Mahmoud Tobias Wagner (advisory role; confirm designation)
Draft Date:	Working draft – September 2026

Declaration page retained for the final thesis.

This is an unsigned working draft. The declaration below must be reviewed and completed by the author for submission.

This is to certify that:

- (i) the thesis comprises only my original work toward the Masters Degree
- (ii) due acknowledgement has been made in the text to all other material used

Mark Emeel Sroor

Date of final submission: to be confirmed

Acknowledgments

Draft note: *Personal acknowledgments will be added by the author in the final version.*

Abstract

The Method of Loci is a memorisation technique in which information is associated with locations along a familiar route. Virtual Reality (VR) can make this process visible and interactive, allowing users to place mnemonic cues in a virtual environment. However, a predefined environment may have little personal meaning to the person using it. A virtual version of a familiar home could provide a more recognisable structure, while also allowing the user to adjust the space to suit the task.

In this thesis, we present a VR memory palace system that supports the creation and furnishing of personalised environments and the placement of visual mnemonic cues. The current study compares a generic palace with a personalised palace based on the participant's home. A within-subjects design is used to examine recall, cognitive load, enjoyment, perceived competence, effort, and presence. The protocol developed through pilot sessions, with changes to word-list difficulty, list length, and the presence questionnaire.

The analysis snapshot available on 22 September 2026 contains 18 complete immediate-recall pairs, 16 pairs at the nominal 24-hour follow-up, and 14 pairs at the nominal one-week follow-up. Immediate mean recall accuracy is 72.39% in the generic condition and 78.67% in the personalised condition. The estimated paired difference is 6.28 percentage points, with a 95% confidence interval from -2.51 to 15.06 percentage points. These values are provisional: the analysis workbook flags six blocking data-quality errors and 130 unreviewed, uncounted responses, and the dataset includes multiple protocol versions. Consequently, this draft does not establish a recall benefit or equivalence between conditions.

The present contribution is the design of the personalised palace workflow and a documented evaluation framework. Content personalisation and auditory mnemonic generation remain part of the broader proposal, but are outside the comparison reported here. The thesis identifies the data reconciliation and experimental controls needed before drawing final conclusions about spatial personalisation.

Contents

Acknowledgments	V
1 Introduction	1
1.1 Motivation	1
1.2 Problem Statement	2
1.3 Objective and Research Questions	2
1.4 Scope and Contributions	3
1.5 Thesis Organisation	3
2 Background and Related Work	5
2.1 Background	5
2.1.1 The Method of Loci	5
2.1.2 Virtual Reality and Presence	5
2.1.3 Familiarity and Personalisation	6
2.1.4 Cognitive Load and User Experience	6
2.2 Virtual Memory Palaces	7
2.2.1 Virtual and Conventional Environments	7
2.2.2 Immersion Aids Recall	7
2.2.3 Explicit Spatial Binding	8
2.2.4 Optimised Palace Design	8
2.3 Alternative Representations and Applications	8
2.3.1 Augmented Reality and Virtual Reality	8
2.3.2 Two-Dimensional Images and Three-Dimensional Objects	9
2.3.3 Learning Beyond Word Lists	9
2.4 Authoring Personalised Environments	10
2.5 Word Properties and Experimental Materials	10
2.6 Position of the Present Study	11
3 Methodology	13
3.1 Study Design	13
3.2 Participants and Apparatus	13
3.3 Pilot Development	14
3.3.1 Task Difficulty	14
3.3.2 Questionnaire Burden	14

3.3.3	Separating Creation from Memorisation	15
3.4	Revised Experimental Procedure	15
3.4.1	Preparation and Training	15
3.4.2	Encoding and Recall	16
3.4.3	Condition Order and List Assignment	16
3.5	Outcome Measures	17
3.5.1	Recall Accuracy and Quantity	17
3.5.2	Recall Order	17
3.5.3	Retention and Recall Duration	18
3.5.4	Questionnaires	18
3.6	Analysis Approach	19
3.7	Data Reconciliation and Validity	19
4	System Design and Implementation	21
4.1	Design Goals	21
4.2	System Overview	22
4.3	Creating the Spatial Environment	22
4.3.1	Home Layout Input	22
4.3.2	Wall Editing	23
4.3.3	Furniture Placement and Persistence	23
4.4	Representing and Placing Mnemonic Content	24
4.4.1	Content Import	24
4.4.2	Image Alternatives	24
4.4.3	Placement and Interaction	24
4.5	Recording the Experiment	25
4.6	Implementation Boundaries	25
5	Preliminary Results and Discussion	27
5.1	Status of the Analysis	27
5.2	Available Observations	27
5.3	Recall Accuracy	28
5.4	Recall Quantity, Order, and Retention	29
5.5	Questionnaire Outcomes	30
5.6	Recall Time	31
5.7	Discussion	31
5.7.1	What the Current Comparison Can Say	31
5.7.2	Familiarity, Construction, and Exposure	32
5.7.3	Mixed Materials and Ceiling Effects	32
5.7.4	Missingness and Order	32
5.7.5	Practical Value and Remaining Evidence	32
6	Conclusion	35
6.1	Achievements	35
6.2	Limitations	35

6.3	Future Work	36
6.4	Closing Remarks	37
Appendix		38
A Draft Completion and Source Record		39
A.1	Purpose of This Draft	39
A.2	Items to Confirm Before Submission	39
A.3	Recorded Protocol Changes	41
A.4	Numerical Reproducibility	41
B Lists		42
	List of Abbreviations	42
	List of Figures	43
	List of Tables	44
References		45

Chapter 1

Introduction

1.1 Motivation

Remembering a list of unrelated words can be difficult. Remembering the way from the front door of one’s home to the kitchen is a different kind of task. The rooms, furniture, and turns along that route already have an order and a meaning. The Method of Loci (MoL) uses this spatial structure to help people remember information that is not itself spatial. The learner associates each item with a location, then mentally follows the route to retrieve the items [1, 2].

For example, a learner could imagine an apple on the front door, a trumpet on the sofa, and a book in the kitchen sink. The unusual images help distinguish the items, while the route provides an order in which to retrieve them. The palace does not have to be a grand building. A house, a familiar room, or another recognisable environment can serve the same purpose. What matters for the technique is the relationship between the information, its cue, and its location.

Virtual Reality (VR) offers a practical way to represent this process. Instead of only imagining a location, the user can enter a virtual environment, select a visual cue, and place it in the space. This makes the operation easier to demonstrate and gives the learner something concrete to interact with. Previous work has investigated immersive displays, active spatial binding, and the design of virtual memory palaces [3, 2, 1]. These studies motivate the use of VR, but they do not answer every question about how a palace should be designed.

Personalisation is one of the open questions identified by that literature. Krokos et al. [3] specifically proposed letting participants build and organise their own virtual palaces and comparing highly personal palaces with ones suitable for a wider group. Yang et al. [13] reported participant interest in personalising a palace used to retrieve knowledge from articles. These are motivations to test personalisation, not evidence that it improves recall. The present study examines one concrete version of the idea: a VR palace based on the participant’s home.

One question concerns the environment itself. A generic apartment can provide rooms and furniture, but it is still a new place for the participant. The learner may have to learn the layout at the same time as learning the material. A palace based on the learner's own home may provide a structure that is already known. Familiarity has been associated with better object recognition and egocentric spatial memory in virtual routes [4]. This gives a reason to investigate familiar layouts, rather than assuming that any form of customisation will improve memory.

Personalisation also has a cost. Drawing walls, arranging furniture, and correcting a layout take time. A system could make users feel more involved while also leaving them tired before the memorisation task begins. Therefore, recall alone is not enough to describe the experience. We also need to consider the effort involved in creating and using the palace, and whether the resulting environment feels understandable, engaging, and present to the user.

1.2 Problem Statement

The problem addressed in this thesis is how to support a personally familiar virtual memory palace and how to evaluate it against a generic palace. A familiar home contains a structure that the user already knows, but turning that structure into a usable VR environment should not require professional modelling skills. The user needs a way to create the walls, furnish the rooms, correct mistakes, and then use the space to associate information with locations.

The comparison must also separate what is being changed from what is being measured. Both conditions in the current study use VR and the MoL. The main difference is whether the environment is generic or personalised. We are therefore investigating the additional contribution of the personalised environment within this workflow. We are not comparing VR with a desktop display, nor comparing the MoL with unaided rehearsal.

1.3 Objective and Research Questions

The objective is to design a VR application that allows users to prepare and use a personalised memory palace, and to investigate the experience and recall outcomes associated with this process. The study is organised around three questions:

1. How does recall in a personalised virtual memory palace compare with recall in a generic virtual memory palace, within the same participants?
2. How do the two conditions compare in cognitive load, enjoyment, perceived competence, effort, and presence?
3. What practical difficulties arise when users create and use a personalised palace, and how do these difficulties affect the interpretation of the comparison?

The first question concerns immediate and delayed recall. The second concerns the participant's experience of the activity. The third concerns the workflow that makes the comparison possible, including preparation time, interface issues, and changes introduced during pilot development. These questions guide the present account; this draft does not claim that their exact wording was preregistered.

1.4 Scope and Contributions

The research proposal covers spatial personalisation, learning-content personalisation, visual mnemonic generation, and auditory mnemonics [5]. Its original title is retained on the title page. The evaluation reported in this draft focuses on the spatial-personalisation component. This narrower comparison is consistent with the supervisory recommendation to evaluate environment and content personalisation separately [6]. Features considered in the wider proposal are not treated as experimentally validated contributions.

The contributions of the present work are the following:

1. A VR memory palace workflow that supports home-based layouts, furniture editing, and the placement of visual mnemonic content.
2. A within-subjects evaluation comparing a generic and a personalised palace, with recall and experience measures.
3. A documented account of pilot-driven changes to the materials and procedure, together with a provisional analysis that makes unresolved data issues explicit.

The personalised condition combines a familiar layout with active construction and additional exposure to that environment. These components are meaningful parts of the implemented experience, but the current comparison cannot determine their separate effects. Accordingly, throughout this thesis, *personalisation* refers to the combined home-based authoring workflow unless a narrower meaning is explicitly stated.

1.5 Thesis Organisation

This thesis is divided into six chapters. Chapter 1 introduces the motivation, problem, and scope. Chapter 2 explains the relevant concepts and reviews related work. Chapter 3 describes the experimental design, pilot development, measures, and analysis approach. Chapter 4 describes the system and the reasoning behind its main interactions. Chapter 5 presents the provisional results and discusses their limitations. Finally, Chapter 6 concludes with the current achievements and the work needed to complete the evaluation. The appendix records outstanding details that must be resolved before submission.

Chapter 2

Background and Related Work

2.1 Background

2.1.1 The Method of Loci

The Method of Loci is a mnemonic technique that associates information with locations. A *locus* is one such location, and *loci* is its plural. The collection of locations is often called a memory palace. To use the technique, the learner chooses a route, associates the information with points along that route, and later follows the same route mentally to retrieve it [1, 2].

There are three parts to this process: the information to be remembered, a representation of that information, and the place where the representation is stored. These parts should not be confused. A chair may be part of the palace and serve as a location. An image of a whale placed on that chair may be the mnemonic representation of the target word *whale*. The chair is not itself the word that the participant is being asked to remember.

The route also provides an ordering structure. A learner who remembers an item but cannot remember where it belongs may succeed at free recall while making an error in ordered recall. Conversely, a learner may correctly preserve the order of a small number of recalled items while forgetting most of the list. This is why recall quantity and recall order are considered separately in the present study.

2.1.2 Virtual Reality and Presence

VR presents a computer-generated environment through which the user can look and interact. A head-mounted display provides a view that changes with the user's head movements, while controllers allow actions such as selecting, moving, and placing objects. For the MoL, this makes it possible to turn an imagined placement into a visible interaction.

Immersion and presence describe related but different aspects of the experience. Immersion concerns the capabilities of the system and the sensory experience it provides. Presence concerns the user's sense of being in the virtual place. A system can offer an immersive display without every user reporting the same level of presence. Similarly, a high presence score does not directly show that the user remembered more information.

Usoh et al. [7] examined presence questionnaires by comparing real and virtual experiences. Their findings caution against treating questionnaire scores as a simple objective measure that transfers unchanged across different kinds of environment. In the present study, both conditions use the same general VR medium. The questionnaire is used to describe the reported experience within that comparison, rather than to establish that the environment was indistinguishable from reality.

2.1.3 Familiarity and Personalisation

Personalisation can mean changing colours, choosing furniture, arranging a room, or recreating a known place. These actions do not necessarily have the same effect on memory. A newly designed room may express the user's preferences while remaining spatially unfamiliar. A simple reconstruction of the user's home may look less elaborate but retain a route that the user already knows.

Merriman et al. [4] compared virtual routes through more and less familiar campus areas. Familiar context benefited novel-object recognition and egocentric direction memory in younger and older adults, but not every spatial measure. A separate group equally familiar with the two areas did not show the same context benefit. The result supports familiarity as a relevant design consideration. It does not demonstrate that editing a virtual room improves verbal recall, which is the more specific question investigated here.

For this thesis, familiarity provides the main rationale for using a home layout. Furniture selection and editing provide a way to make that layout useful and recognisable. They also introduce agency and extra exposure, which must be considered when interpreting the personalised condition.

2.1.4 Cognitive Load and User Experience

The user must carry out several activities during a VR memory task. These include understanding the controls, navigating the environment, choosing a mnemonic image, finding a location, and learning the association. Some of this effort belongs to the memory task itself. Some comes from the way the application presents that task.

The current questionnaire separates intrinsic, germane, and extraneous cognitive-load scores. In this draft, these names refer to the operational subscales used in the study workbook, rather than proof that three independent cognitive processes have been isolated. The precise source and adaptation of the administered cognitive-load instrument

still require confirmation. In particular, the workbook preserves a two-item germane score and a separate three-item exploratory variant [8].

The Intrinsic Motivation Inventory (IMI) provides another view of the experience. Its interest/enjoyment, perceived competence, and effort/importance subscales address different questions [9]. A participant can enjoy the activity while finding it difficult. They may also feel competent without recalling more words. These measures therefore complement recall performance; they do not replace it.

2.2 Virtual Memory Palaces

2.2.1 Virtual and Conventional Environments

Legge et al. [10] investigated whether a recently learned virtual environment could support the MoL, compared with a conventional environment. Their findings showed that useful memory-palace performance did not necessarily require a long-established real-world location. This matters because it prevents an overly simple assumption that a generic virtual palace cannot work. A new environment can provide sufficient structure after familiarisation.

The present study asks a different question. Both conditions provide a virtual palace, and both allow the participant to use the same general mnemonic technique. The question is whether reconstructing a personally familiar environment offers an additional advantage over the generic one under the practical conditions of the study.

2.2.2 Immersion Aids Recall

Krokos et al. [3] compared a head-mounted display with a desktop display in a memory-palace task involving faces and names. Forty participants took part. Mean recall accuracy was 84.05% in the head-mounted-display condition and 75.24% in the desktop condition. The absolute difference is approximately 8.81 percentage points.

This study supports the possibility that the display and embodied viewing experience can affect recall. However, its manipulation was the display condition, not whether the palace represented the participant's home. Its task also differed from freely placing word cues throughout rooms. The result motivates using VR as a platform, but it cannot be used as an estimate of the effect expected from spatial personalisation.

In their future-work discussion, Krokos et al. also called for studies in which users build, manipulate, and organise their own palaces, and for a comparison between highly personal and more widely usable palaces [3]. Their experiment did not test those proposals. The distinction matters here: the present generic-versus-home comparison addresses the personal-space question, while active authoring and familiarity remain combined in its personalised condition.

2.2.3 Explicit Spatial Binding

Reggente et al. [2] studied the role of actively binding objects to locations. Participants experienced three virtual environments, each containing a sequence of 15 objects. The experimental group could lock objects into place, while the control group was not given that interaction. Each group contained 30 participants. The experimental group recalled 28% more objects, despite matched exposure duration and instructions.

The relevance of this work lies in the association between the item and its location. A memory-palace application should not simply display pictures inside a room. It should support the user's deliberate act of placing a cue somewhere meaningful. This informed the present system's emphasis on placement. At the same time, active placement and environment familiarity are different factors. Evidence for one does not establish the effect of the other.

2.2.4 Optimised Palace Design

Moll and Sykes [1] provide the main procedural reference for this project. Their feasibility study developed a VR memory palace around design choices intended to support immersion, imagery, and the learning of the MoL. Participants could associate words with images and use an apartment-like environment for memorisation.

In the first week, their reported mean recall scores increased from 62.55 to 82.91. In the second week, the reported means increased from 63.44 to 85.67. These findings support the feasibility of teaching and using the technique through a purpose-built VR application. However, they concern a comparison with traditional memorisation and repeated use of their system. They should not be treated as the expected gain from allowing a participant to recreate their home.

The paper also discusses familiarity, distinct locations, and the value of meaningful imagery. These considerations are useful for the present design: rooms should be distinguishable, cues should have stable locations, and the participant should understand why each image represents its word. The present study extends this design direction by making the spatial environment itself personal to the participant.

Moll and Sykes let participants choose among images for each word and suggest that allowing an unrestricted personal image choice could be investigated further [1]. That is personalisation of the mnemonic cue, whereas this thesis varies the spatial setting. Separating those choices helps identify what a later experiment would need to control.

2.3 Alternative Representations and Applications

2.3.1 Augmented Reality and Virtual Reality

Caluya et al. [11] compared AR and VR memory palaces for second-language vocabulary acquisition, with a paper condition as a reference. Their exploratory results did not

demonstrate that the AR or VR conditions outperformed the paper condition. They also examined how participants relied on prior vocabulary and semantic associations.

This is useful counterevidence to the assumption that adding immersive technology automatically improves learning. A participant’s strategy, the material, and the fit between training and testing all matter. In the present project, VR is chosen because the palace can be reconstructed, edited, and accessed without requiring the user to be physically inside the original home. That practical reason is distinct from claiming that VR is generally superior to AR for memory.

2.3.2 Two-Dimensional Images and Three-Dimensional Objects

Sandberg Bröms [12] compared 2D images with 3D objects in a mixed-reality MoL application. The between-subjects study involved 20 participants and did not find a statistically significant difference in retention between the representations. The thesis also reported limitations concerning sample size, physical space, and the number of objects.

This does not establish that the two representations are equivalent. It does, however, give no basis for assuming that every target word needs a custom 3D model. Images provide a practical representation for a larger range of words, while furniture and walls can still provide the spatial structure of the palace. The present study uses visual cues within that structure; it does not independently test image dimensionality.

2.3.3 Learning Beyond Word Lists

Yang et al. [13] extended the memory-palace idea to knowledge retrieved from scholarly abstracts. They compared a baseline condition, an image-based palace variant, and a VR-based variant. The VR variant improved retrieval relative to the baseline and showed a more moderate improvement relative to the image-based variant. Anecdotal participant feedback suggested that personalising a palace would be welcome; the study did not test that preference as a recall intervention.

This work shows why memory-palace research need not end with isolated words. Nevertheless, remembering words and understanding an article are different outcomes. Word lists are useful for the present controlled comparison because each target can be identified and scored. Any claim that the system improves academic understanding would require a separate task that measures comprehension and transfer.

Reddy’s design thesis [14] explores AI-generated cues and AR through research-through-design and speculative design. It contributes a perspective on how memory support might be designed, rather than a controlled estimate of the benefit of personalised VR palaces. It is therefore relevant to the broader direction of this project, while remaining distinct from evidence for the current experimental comparison.

2.4 Authoring Personalised Environments

The ability to edit an environment introduces an interaction-design problem as well as a memory problem. Full automation can reduce manual work, but the generated result may not correspond to what the user intended. For a familiar home, the user knows whether a doorway is on the wrong wall or whether two rooms have been connected incorrectly.

Perera et al. [?] described a customisable mobile VR memory-palace prototype and proposed offering different palace templates according to user preference. This supports personalisation as a system-design direction, but their short conference abstract does not establish a recall benefit from preference-based templates. A home reconstruction is also more specific than choosing a preferred generic template: it brings prior knowledge of the route into the virtual environment.

VRCopilot, developed by Zhang et al. [15], examines generative AI as part of immersive layout authoring. It compares manual, scaffolded, and automatic creation. Its studies found that scaffolded creation supported greater agency than automatic creation, while manual creation produced the highest reported creativity and agency. These findings concern authoring experience, not memory performance.

The design implication for this project is to keep generated or imported layouts editable. AI assistance can help translate an input into a starting layout, but the participant should be able to correct the result. A technically valid floor plan is not enough if it fails to preserve the structure that makes the home familiar.

2.5 Word Properties and Experimental Materials

The words used in a memory experiment are not interchangeable. Imageability, frequency, and other properties can affect performance. Madan, Glaholt, and Caplan [16] showed that item properties can influence associative memory in different ways. In their paired-associate experiments, imageability was particularly related to association retrieval, while frequency was more strongly related to target-item recall.

Madan’s later memorability work [17] examined item free-recall probabilities for a large word set. Such estimates provide a way to organise candidate words by observed memorability and examine other lexical properties. They are useful for constructing lists, but the estimated probabilities come from a different task. They are not calibrated predictions of recall after a participant has selected images and placed them in VR.

This distinction became important during pilot development. Lists that appeared suitably balanced could still be too easy when combined with memorable images and spatial associations. List length, memorability, and image choices therefore need to be recorded as part of the experimental materials, rather than assuming that a single label such as *hard* fully describes the task.

2.6 Position of the Present Study

Table 2.1 summarises the main comparisons that inform this thesis. The studies support different parts of the design. Some concern the display, others concern active placement, and others concern authoring or the representation of information. None of these factors should be substituted for another when explaining the results.

Table 2.1: Selected related work and its relationship to the current study.

Work	Main comparison or contribution	Relevance and boundary
Legge et al. [10]	Virtual versus conventional palace environments	A newly learned palace can work; does not isolate home reconstruction.
Krokos et al. [3]	Head-mounted versus desktop display	Motivates immersion; does not test personalisation.
Perera et al. [?]	Customisable mobile VR palace	Proposes preference-based templates; no tested home-layout effect.
Reggente et al. [2]	Active spatial binding versus a control	Motivates deliberate placement of cues.
Moll and Sykes [1]	Optimised VR MoL feasibility	Informs palace design and study procedure.
Merriman et al. [4]	More versus less familiar context	Supports familiarity rationale, with different memory outcomes.
Zhang et al. [15]	Manual, scaffolded, and automatic VR authoring	Informs editable assistance; does not measure recall.
Present study	Generic versus home-based personalised VR palace	Tests the combined familiar-layout and authoring workflow.

The present work brings these considerations together in one comparison: participants use both a generic palace and a palace based on their home. It examines whether the personalisation workflow is associated with different recall and experience outcomes. The comparison remains a test of the workflow as a whole, because familiarity, editing, and additional environmental exposure have not been independently manipulated.

Chapter 3

Methodology

3.1 Study Design

The study compares two ways of using the same general VR memory-palace workflow. In the generic condition, referred to as Condition A in the protocol notes, the participant uses a predefined environment. In the personalised condition, referred to as Condition B, the participant uses an environment based on their home. Both conditions involve placing and memorising word cues.

A within-subjects design was used so that each participant could serve as their own comparison. This reduces the extent to which differences in general memory ability explain differences between conditions. However, it also introduces possible practice, fatigue, and carry-over effects. The procedure therefore planned to vary condition order and word-list assignment across participants.

The account in this chapter draws on the local methodology notes, dated work and meeting logs, and the structure of the current analysis workbook [18, 8]. These sources describe different stages of the study. The early protocol page lists 30 words and 15 minutes of memorisation, while the local revised protocol lists 25 words and 10 minutes. We report this development explicitly instead of presenting the revised procedure as if it applied to every collected session.

3.2 Participants and Apparatus

The current workbook contains complete immediate-recall data for 18 paired participants. This is an analysis count, not a verified statement of the total number recruited. Follow-up and questionnaire counts differ because some outcomes are missing or unresolved. Recruitment method, demographic distribution, eligibility criteria, compensation, withdrawals, and the separation between pilot and main-study participants have not yet been reconciled in the materials used for this draft.

The application was developed in Unity, as documented in the January supervisory meeting, and used a VR headset and controllers [6]. The available work logs describe builds for the headset, but the exact headset model and study-build version are not specified in the fetched proposal. These details should be recorded from the actual study equipment rather than inferred from the devices used by reference papers or the earlier bachelor’s project.

Draft note: *Before submission, add the recruitment and demographic description, hardware model, Unity and application versions, locomotion configuration, study location, consent procedure, and ethics approval or applicable institutional review status. No approval identifier is asserted in this draft.*

3.3 Pilot Development

3.3.1 Task Difficulty

The local methodology notes report that an early pilot used 30 highly imageable words and produced perfect immediate recall. Such a result is useful for checking whether the participant understands the task, but provides little room to detect improvement. The materials were subsequently revised using word memorability estimates.

The work log dated 17 July 2026 describes a 20-word arrangement with five easy, ten moderate, and five hard words [19]. The 13 August log then describes two 25-word lists, each containing ten easy, ten medium, and five hard words. The reported mean memorability estimates were 0.742430 and 0.742425. The same entry states that architecture and location words were excluded and that lists were balanced on memorability, concreteness, word length, syllables, and frequency [20].

Later local notes describe a move to 25 low-memorability words. The current workbook contains three named sets: *Condition List*, *Memory List*, and *Hard Memory List*. Its included recall records use 20 targets for the Condition List and 25 for the other two sets. This confirms that the snapshot is not restricted to one final materials version.

The distinction matters for the analysis. Dividing correct responses by list length makes the scores comparable as proportions, but does not remove differences in difficulty. A shorter list with easier words is still a different task. The final study report should identify the intended main-study cohort and show whether conclusions change when earlier versions are excluded.

3.3.2 Questionnaire Burden

The initial protocol included the Igroup Presence Questionnaire (IPQ). The 10 August meeting notes recommend a shorter presence questionnaire and switching the word lists

between conditions across tests [21]. The 12 August work-log title records the change to the Slater–Usch–Steed (SUS) presence questionnaire [22].

Here, SUS means the six-item presence measure, not the System Usability Scale. The distinction is necessary because both instruments use the same abbreviation. The workbook retains legacy IPQ fields separately and flags unresolved endpoint scoring. The two instruments are not combined into one presence score.

3.3.3 Separating Creation from Memorisation

The 27 July meeting notes place customisation before both experimental conditions, with a break and cognitive-load assessment after customisation [23]. This separates the effort of creating the palace from the effort of memorising the words. It also means that the participant has already spent time inside or working on the personalised environment before the condition comparison begins.

This arrangement helps avoid placing the entire authoring task between the two memory conditions. Nevertheless, it does not equalise environmental exposure. The generic palace is encountered during training, while the personalised palace receives additional attention during construction. That difference is retained as a limitation of the workflow comparison.

3.4 Revised Experimental Procedure

Figure 3.1 summarises the revised procedure recorded in the local notes. The timing below describes that version; per-session deviations must be checked against session records.

3.4.1 Preparation and Training

Before attending, participants were asked to draw a simple wall layout of their home. The local protocol describes a custom ChatGPT workflow that converted this drawing into a CSV file for import. Participants could review and correct the imported layout. This step was intended to reduce the effort of building the environment from an empty scene.

At the start of the session, the participant received an explanation of the MoL and an opportunity to ask questions. Training then introduced the VR controls, the generic environment, and cue placement. A short five-word task allowed the participant to practise the technique before the main memorisation task.

The participant then prepared the personalised palace, correcting the walls where necessary and furnishing the rooms. The local notes recommend approximately five furniture items per room to keep this stage manageable and encourage room-based grouping. This was a preparation guideline, not a claim that every room contained exactly five items. After saving the environment, the participant took a short break and completed the first cognitive-load questionnaire.

3.4.2 Encoding and Recall

For each condition in the revised procedure, the participant had ten minutes to place and memorise the assigned 25-word list. The participant then completed questionnaires covering cognitive load, enjoyment, perceived competence, and presence before recalling the words. The local notes specify up to five minutes for verbal recall.

The workbook labels this first recall as *Immediate*. In procedural terms it follows the questionnaires, so it is not a zero-delay test. The exact interval between encoding and recall may vary with questionnaire completion time. This should be reported when session timing has been reconstructed.

After a short break, the participant completed the other condition with the other list. Follow-up records in the workbook are labelled *Delayed24h* and *Delayed1Week*. These are nominal categories. Their labels alone do not establish that each test occurred at precisely 24 hours or seven days. The final report should specify follow-up delivery, permitted timing windows, and whether any rehearsal or access to cues was controlled.

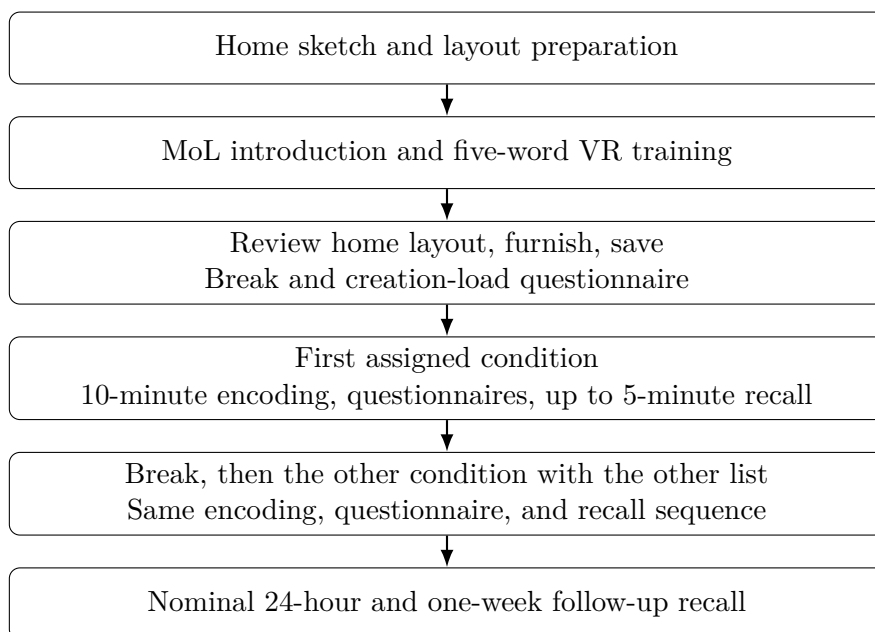


Figure 3.1: Revised procedure reconstructed from the local protocol and supervisory notes. Earlier sessions used different materials; actual follow-up timing remains to be verified.

3.4.3 Condition Order and List Assignment

Condition order and list assignment are separate design factors. Reversing the order of the conditions does not, by itself, ensure that each word list is used equally often in each

Table 3.1: Intended counterbalancing structure, not verified allocation counts. G denotes generic and P denotes personalised.

Schedule	First condition	Second condition
1	G, List 1	P, List 2
2	G, List 2	P, List 1
3	P, List 1	G, List 2
4	P, List 2	G, List 1

condition. Table 3.1 gives the four schedules implied by crossing two condition orders with two list assignments.

The original local prose describes four combinations but abbreviates them ambiguously. The workbook’s session-order diagnostic currently has zero complete pairs. Consequently, the table expresses the intended design and does not claim that those schedules were achieved in balanced numbers. Condition codes, list identifiers, and actual presentation order need to be reconciled independently before order effects can be evaluated.

3.5 Outcome Measures

3.5.1 Recall Accuracy and Quantity

Let C_{ict} be the number of unique correctly recalled targets for participant i , condition c , and timepoint t , and let L_{ict} be that trial’s target-list length. Recall accuracy is

$$A_{ict} = \frac{C_{ict}}{L_{ict}}. \quad (3.1)$$

A target contributes at most once to C_{ict} . Repeating a correct word does not increase the score. Raw quantity is also retained because it expresses the result in words, but it should not be interpreted without considering the list length.

The workbook distinguishes omissions, intrusions, repetitions, and responses awaiting review. Its reported number of words recalled is retained for audit, while the score is computed from the responses and the resolved target list. Unapproved near matches are not silently counted as correct. Missing trials are left missing rather than scored as zero [8].

3.5.2 Recall Order

The analysis includes the length of the longest common subsequence (LCS) between the target order and the sequence of first matched responses. Dividing by target-list length gives

$$O_{ict} = \frac{\text{LCS}(\text{target sequence, matched response sequence})}{L_{ict}}. \quad (3.2)$$

This measures how much of the original sequence was recovered in order while allowing omissions. For example, for targets A, B, C, D, responses A, C, D yield an LCS of three. Responses D, C, A contain the same three items but preserve less of the target sequence.

Kendall's order measure is also recorded for recalled items. It describes relative ordering within that recalled subset. It should be considered alongside accuracy, because correctly ordering a small subset is not the same as remembering the full list. Undefined order statistics remain missing.

3.5.3 Retention and Recall Duration

Retention change is calculated as later accuracy minus immediate accuracy for the same participant, condition, list set, and ordered targets. A negative value indicates lower recall at the follow-up. This calculation uses matched longitudinal records rather than subtracting group means from different participant subsets.

Recall duration is available only when an explicit valid duration is present and the trial is marked as timed. The workbook does not infer duration from arbitrary timestamps for untimed trials. Longer recall can mean that a participant continued searching, recovered more items, or struggled with retrieval. It is therefore not automatically classified as a worse outcome.

3.5.4 Questionnaires

Table 3.2 summarises the main scoring definitions in the workbook. Items are represented on a seven-point scale. Where signed responses were used, the workbook first converts them using $x + 4$; reverse-scored items then use $8 - x$. A subscale is computed only when all required items are valid [8].

Table 3.2: Questionnaire scoring definitions in the current analysis workbook.

Measure	Items	Scoring detail
IMI interest/enjoyment	7	Mean; items 3 and 4 reversed.
IMI perceived competence	6	Mean; item 6 reversed.
IMI effort/importance	5	Mean; items 2 and 5 reversed.
SUS presence	6	Mean; distinct from legacy IPQ and the System Usability Scale.
Intrinsic cognitive load	2	Mean of the two recorded items.
Germane cognitive load	2	Main definition; three-item variant kept exploratory.
Extraneous cognitive load	3	Mean of the three recorded items.

These definitions describe how the present workbook produces scores. They do not replace verification against the exact questionnaire forms administered. Internal consistency estimates and the separate creation-load results are not reported in this draft because they have not been verified from a reconciled item-level dataset.

3.6 Analysis Approach

For each outcome, the comparison uses participants with a valid score in both conditions. The paired difference is defined consistently as

$$d_i = X_{i,\text{personalised}} - X_{i,\text{generic}}. \quad (3.3)$$

The workbook calculates the mean difference $\bar{d}$, the standard deviation of differences s_d , and a paired t statistic:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad d_z = \frac{\bar{d}}{s_d}. \quad (3.4)$$

The 95% confidence interval uses the t distribution with $n - 1$ degrees of freedom. The paired effect size d_z uses the variability of within-participant differences, not the pooled standard deviation of the two condition scores.

The workbook also provides Wilcoxon signed-rank results. Its dictionary specifies conditional sign permutation for up to 50 nonzero differences, with tied midranks and zero differences omitted only from that signed-rank calculation. Zero differences remain in the descriptive paired sample. This is useful because the recall scores are bounded and some participants have identical scores in the two conditions.

The workbook labels immediate accuracy as primary and 24-hour accuracy as the primary follow-up. These are reporting priorities in the analysis file; a dated preregistration was not established from the available sources. Other recall, questionnaire, and exploratory outcomes should not be searched selectively for favourable results. The present draft reports the stored statistics as unadjusted provisional outputs rather than presenting them as a final confirmatory test.

3.7 Data Reconciliation and Validity

The snapshot's run is marked complete, but its interpretation gate is blocked. Completing calculations and approving the underlying records are different steps. Six blocking errors and 130 unreviewed, uncounted responses remain. The visible error rows include duplicated eligible questionnaire records. An error count is not a count of affected participants, because one unresolved record can affect several subscales.

Before final analysis, the records need to establish participant identity, condition, list version, session order, target sequence, timepoint, and the retained response record. The final analysis should also distinguish main-study sessions from pilot sessions. Participant names have been omitted from the thesis tables and the local numerical snapshot uses pseudonymous identifiers.

The principal threats to validity are unequal exposure to the environments, changes in materials across sessions, missing follow-up outcomes, unresolved scoring decisions, and an incomplete reconstruction of condition order. These limits are not resolved by a paired statistical test. They determine what the test can reasonably be said to compare.

Chapter 4

System Design and Implementation

4.1 Design Goals

The system was designed to let users create a memory palace without first learning a general-purpose 3D modelling application. The required operations follow the memory task: prepare a space, identify useful locations, associate content with those locations, and revisit the associations.

The first design goal was to preserve recognisable spatial structure. For a home-based palace, the connection between rooms can be more useful than fine visual detail. A familiar doorway in the wrong place changes the route that the participant expects to follow. Layout editing therefore has a direct role in the memory task, rather than being only a cosmetic feature.

The second goal was to make placement deliberate and understandable. The user should be able to choose an image, relate it to a word, and place it at a location. This follows the emphasis on explicit spatial binding in previous MoL research [2]. The interaction should make the association clear without requiring complicated controller operations.

The third goal was to allow revision. Imported layouts and selected furniture may need correction, and a placed cue may need to be moved. The work logs document iterative improvements to wall editing, furniture movement, rotation, deletion, and menu placement [24, 19]. These operations allow users to recover from mistakes during preparation.

Finally, the system needed to support an experiment as well as an individual learning session. Conditions, word lists, saved environments, and recorded results must remain identifiable. A visually successful session is insufficient if its recall file cannot later be linked to the correct target list.

4.2 System Overview

The prototype is implemented in Unity [6]. Figure 4.1 shows the functional workflow reconstructed from the development logs. It is a description of the main data and interaction stages, not a class diagram or an assertion about uninspected source-code modules.

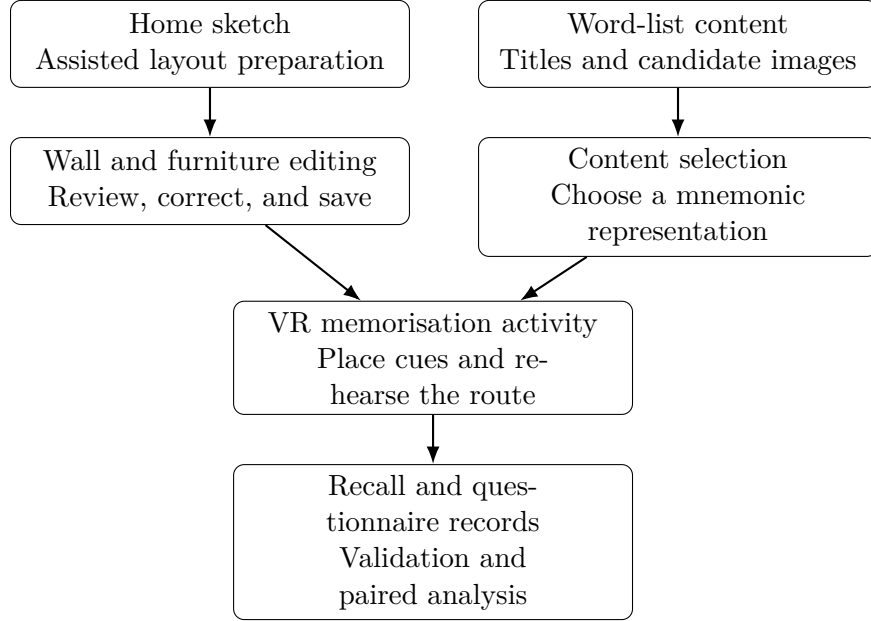


Figure 4.1: Functional overview of the documented workflow. Analysis includes records collected outside the VR application.

The spatial environment and the learning content play different roles. Walls and furniture define the palace. Word cues define the current material. Keeping these concepts separate makes it possible, in principle, to use a saved palace for a different list without rebuilding the home. Whether repeated reuse introduces interference between lists is a separate question for evaluation.

4.3 Creating the Spatial Environment

4.3.1 Home Layout Input

The revised protocol begins with a participant’s sketch of their home. The local notes describe using ChatGPT to convert that sketch into a CSV representation that can be imported into the application. Earlier work logs also record experimentation with mapping a board to walls and with assisted room and furniture placement [25, 26].

The purpose of assistance is to reduce the manual work needed to reach an editable starting point. It does not remove the need for review. A sketch can leave room dimensions ambiguous, omit door openings, or contain lines that are not walls. The participant’s

inspection is therefore part of the preparation workflow: the reconstructed space must correspond to the place they intend to use.

The exact layout-file schema and the conversion workflow used by each study build remain to be documented from the implementation. The CSV description in the protocol refers to layout input, whereas the JSON example in the April development log refers to mnemonic content. These should not be presented as contradictory descriptions of one file type.

4.3.2 Wall Editing

The July development log documents extensions to the wall maker, including drawing and editing operations, moving walls, a room operation, and a 3D camera view [19]. These features address a basic problem with reconstructed spaces: users need to correct the geometry in terms they understand.

A room-based operation can make it easier to express a large part of the floor plan, while individual wall operations allow local corrections. Viewing the result in three dimensions helps connect the drawn layout with the environment that the user will later navigate. The goal is a recognisable and usable palace, rather than an architecturally precise digital model.

This distinction also affects measurement. The workbook's layout metrics use grid units and graph structure. They should not be reported as floor area, metres travelled, or other physical quantities without a verified scale and corresponding measurements [8].

4.3.3 Furniture Placement and Persistence

The December development log records a furniture menu, editing of placed furniture, and saving and loading both furniture placement and wall layout [24]. The documented edit operations include moving, rotating, and deleting items.

Furniture provides landmarks within otherwise similar rooms. It also gives the participant places at which to position mnemonic cues. The revised study guidance of approximately five items per room attempts to balance recognisability with preparation effort. More furniture is not automatically better: it can increase the number of distinct locations, but can also make placement and navigation more difficult.

Saving the environment allows the authoring stage to end before the memory comparison begins. It also provides a record of the environment used. For reproducibility, the saved layout should be linked to the participant and condition in the session data. An environment that cannot be linked unambiguously to a session cannot safely be used in participant-level geometry analyses.

4.4 Representing and Placing Mnemonic Content

4.4.1 Content Import

The 13 April log documents a scrollable menu that imports entries from JSON and loads associated images [27]. The recorded entry structure includes an image filename, a title, and a description. The following simplified example follows that documented structure:

```
{
  "entries": [
    {
      "imageFile": "stack.png",
      "title": "Stack",
      "description": "A last-in, first-out data structure."
    }
  ]
}
```

The example demonstrates that the content representation can hold a concept and a short explanation. It does not demonstrate that learning computer-science concepts was evaluated in the present experiment. The reported study uses word lists.

4.4.2 Image Alternatives

The July log describes generating three distinct ideas per word so that the image options do not all depict the same interpretation. The August log likewise records three picture-search queries for each word [19, 20]. The documented workflow therefore supports variety in candidate imagery.

This should not be described as proof that the final images were generated by an image model. Generating search ideas, retrieving pictures, and synthesising pictures are different operations. The exact image provider, selection rules, and licensing information should be added from the final study build.

Offering alternatives gives the participant a chance to choose a cue that makes sense to them. That choice is relevant because the cue should support an association with the target word. An attractive image is not useful if the participant interprets it as a different object. Ambiguous cues can also complicate recall scoring when a response describes the selected picture rather than matching the target word exactly.

4.4.3 Placement and Interaction

The development logs document work on locus placement and subsequent revisions to that interaction [28, 29]. Together with the study protocol, they establish that selecting

and placing cues is part of the user task. The precise controller mappings, attachment behaviour, and locus ordering should be confirmed against the actual study build before a final implementation specification is written.

Menu positioning was one practical concern. The July log reports moving the menu closer to the player and reducing its size because it could be hidden behind furniture [19]. This matters to the experimental task: time spent finding a hidden menu is time unavailable for building associations. Interface difficulty can also raise cognitive-load ratings independently of palace familiarity.

No quantitative usability improvement is claimed from this change. The log establishes that the issue was identified and addressed during development, not that a controlled comparison measured the effect of the fix.

4.5 Recording the Experiment

The analysis workbook combines recall records, questionnaire data, and Quest event metrics [8]. Recall records contain condition and list information, response sequences, nominal timepoints, target references, and, for timed trials, explicit duration fields. These fields make it possible to inspect how a score was produced rather than relying only on a manually entered total.

The dictionary distinguishes event-based mode durations from cumulative totals. Cumulative timing rows are retained for audit but excluded from condition timing totals. It also notes that transition-spanning intervals can be assigned to the condition active at the end of the mode. This is a material limitation if timing is later used to explain differences between conditions.

The same caution applies to geometric metrics. A straight-line route joining loci is not the distance the participant actually travelled. Wall-length totals in grid units are not measurements in metres. The present results therefore concentrate on the directly defined recall and questionnaire outcomes; they do not infer navigation efficiency from those proxy measures.

4.6 Implementation Boundaries

The documented system supports the central steps needed for the current study: environment preparation, correction, furnishing, saving, content selection, and spatial placement. The evidence available for this chapter consists of the protocol and development records. The thesis folder does not contain the Unity source code or final application screenshots, so exact algorithms, class names, software versions, and performance figures are not invented here.

The wider proposal also includes content segmentation, generated visual mnemonics, and auditory cues. Some development records demonstrate content-import capability

and exploration of generative tools. However, a planned or prototyped feature is not the same as a feature used in every experimental session. The present evaluation concerns the spatial-personalisation workflow. Separate studies would be needed to determine the contribution of the other proposed components.

Chapter 5

Preliminary Results and Discussion

5.1 Status of the Analysis

This chapter reports the workbook snapshot retrieved on 22 September 2026 [8]. It is a provisional description of the available outputs, not an approved final analysis. The workbook marks its calculation run as complete, but explicitly marks the data as unsafe to interpret because six blocking data-quality errors remain. It also lists 130 unreviewed, uncounted responses.

The distinction affects every result below. A stored mean and confidence interval can be arithmetically correct while relying on records whose scoring or inclusion still needs review. The visible blocking errors include duplicate eligible questionnaire records. Legacy presence scoring, unresolved responses, mixed list versions, and unavailable order diagnostics create further limitations.

For this draft, the reported recall means, paired differences, t statistics, and d_z values were checked against the workbook’s included trial-level accuracy values. They agree within rounding tolerance. This verifies the transcription and arithmetic of the snapshot, not the correctness of all source decisions. A pseudonymous local snapshot and a small verification script accompany the thesis source.

5.2 Available Observations

Table 5.1 reports complete-pair counts for the main outcomes. The number of participants differs by measure. A missing outcome was not replaced with zero, and the maximum number of available recall pairs was not reused as the sample size for every questionnaire.

The included recall records comprise 36 immediate trials, 32 nominal 24-hour trials, and 28 nominal one-week trials. The 36 immediate records include eight 20-target Condition List trials, twelve 25-target Memory List trials, and sixteen 25-target Hard Memory List trials. Thus, the immediate comparison pools more than one materials version.

Table 5.1: Complete pairs available in the provisional workbook snapshot.

Outcome	Complete pairs
Immediate recall accuracy	18
Nominal 24-hour recall accuracy	16
Nominal one-week recall accuracy	14
Immediate recall duration	14
IMI subscales	17
SUS presence	14
Cognitive-load subscales	18
Legacy IPQ comparison	0
Session-order diagnostic	0

These counts do not establish a final recruitment flow. Some sessions may belong to pilot development, and the participant registry still requires review. The final report needs a separate account of recruitment, exclusions, main-study eligibility, and missing outcomes.

5.3 Recall Accuracy

Table 5.2 gives the condition means and standard deviations for accuracy. Each row uses the same complete paired subset for its two condition means. Different timepoints can involve different subsets.

Table 5.2: Provisional recall accuracy as a percentage of the trial’s target list. SD denotes sample standard deviation.

Timepoint	n	Generic mean (SD)	Personalised mean (SD)
Immediate	18	72.39 (26.21)	78.67 (24.46)
24 hours	16	59.63 (29.58)	66.13 (29.71)
One week	14	52.07 (32.57)	57.43 (30.89)

Immediate accuracy is numerically higher in the personalised condition by 6.28 percentage points. The workbook’s paired estimate has a 95% confidence interval from -2.51 to 15.06 percentage points, $t(17) = 1.508$, $p = .150$, and $d_z = 0.355$. Eight participants have higher personalised scores, seven have higher generic scores, and three have equal scores. The median paired difference is zero.

The mean difference therefore does not describe a uniform benefit across participants. It also does not establish equivalence: the interval includes both a negative difference and a potentially meaningful positive difference. Even if the underlying records were

fully reconciled, the uncertainty would remain important. In the present snapshot, the data-quality block provides an additional reason to withhold a final conclusion.

At the nominal 24-hour follow-up, the paired difference is 6.50 percentage points, with a 95% interval from -4.33 to 17.33 . At one week, it is 5.36 percentage points, with an interval from -1.26 to 11.97 . Table 5.3 reports the corresponding stored statistics.

Table 5.3: Provisional paired recall comparisons. Differences and confidence intervals are in percentage points, personalised minus generic. All p values shown are unadjusted.

Timepoint	Difference	95% CI	p_t	d_z	p_W
Immediate	6.28	$[-2.51, 15.06]$	.150	0.355	.425
24 hours	6.50	$[-4.33, 17.33]$	.220	0.320	.196
One week	5.36	$[-1.26, 11.97]$	.104	0.467	.236

The signed-rank results do not remove the uncertainty apparent in the mean-based comparisons. Their different values also show why selecting whichever test gives the smallest p value would be misleading. The result of interest is the paired outcome and its uncertainty, interpreted in the context of the design and data quality.

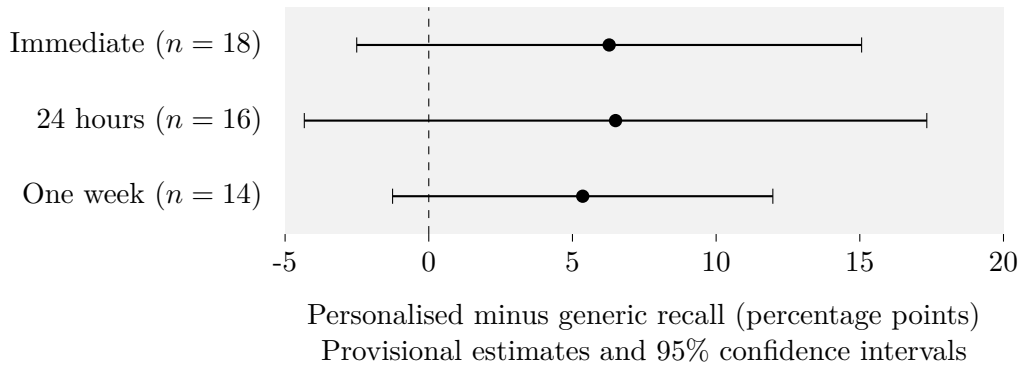


Figure 5.1: Paired recall differences in the frozen snapshot. Every interval crosses zero. The workbook remains blocked for interpretation pending data review.

5.4 Recall Quantity, Order, and Retention

Immediate raw recall quantity averages 17.28 words in the generic condition and 18.89 in the personalised condition, giving a paired difference of 1.61 words. These means pool 20- and 25-target lists. It would therefore be incorrect to divide the overall word-count mean by 25 and expect it to reproduce the mean of trial-level accuracy.

The mean LCS proportion is 0.681 for the generic condition and 0.735 for the personalised condition at immediate recall. The paired difference is 0.054, with a 95% interval

from -0.034 to 0.143 and an unadjusted $p = .210$. The direction resembles the accuracy comparison, but the interval remains broad.

Kendall recalled-order means are 0.870 and 0.887 at immediate recall. At the nominal one-week follow-up, they are 0.872 and 0.708 , using 13 complete pairs. This contrast illustrates why quantity and order should not be collapsed into a single claim. More recalled items do not necessarily mean better ordering among those items, and the set of recallable items changes over time.

For the 16 complete 24-hour retention pairs, mean accuracy change from immediate recall is -0.145 in the generic condition and -0.159 in the personalised condition. The paired difference in change is -0.014 , with a 95% interval from -0.098 to 0.069 . The corresponding one-week changes are -0.254 and -0.245 for 14 pairs. These are within-participant retention calculations from the workbook. They are not obtained by subtracting the rows of Table 5.2, whose participant subsets differ.

The snapshot therefore does not support a claim that personalisation slows forgetting. Numerically higher delayed accuracy can arise alongside similar losses from immediate recall. A final retention analysis should preserve the same participants, target lists, and scoring rules across the compared timepoints.

5.5 Questionnaire Outcomes

Table 5.4 reports condition means for the main questionnaire subscales. Scores are on the workbook's seven-point scale. The signs of the differences must be interpreted according to the construct: more enjoyment and more extraneous load are not favourable in the same way.

Table 5.4: Provisional questionnaire results. G denotes generic and P personalised; $\Delta = P - G$. Confidence intervals and p values concern paired differences and are unadjusted.

Measure	n	G	P	Δ	95% CI	p
Interest/enjoyment	17	5.85	6.02	0.17	$[-0.18, 0.52]$	.326
Perceived competence	17	4.46	4.77	0.31	$[-0.49, 1.11]$	.418
Effort/importance	17	5.58	5.39	-0.19	$[-0.52, 0.15]$	.252
SUS presence	14	5.17	5.48	0.31	$[-0.03, 0.65]$	.069
Intrinsic load	18	4.61	4.19	-0.42	$[-1.34, 0.51]$	.356
Germane load	18	3.86	3.92	0.06	$[-0.60, 0.72]$	.861
Extraneous load	18	3.13	2.69	-0.44	$[-1.04, 0.15]$	.134

Enjoyment is high in both conditions, with a small positive personalised-minus-generic difference. Perceived competence and presence are also numerically higher in the personalised condition. Intrinsic and extraneous load are numerically lower. However, every confidence interval shown includes zero, and the duplicate-record issues directly concern questionnaire data.

The presence result should not be described as significant or nearly proven because its unadjusted p value is .069. It is one of several outcomes, uses only 14 pairs, and remains subject to source reconciliation. Nor does it establish a mechanism by which personalisation improves recall. Such a claim would require an appropriate analysis of the relationship between presence and recall, as well as stronger control over alternative explanations.

Effort/importance is slightly lower in the personalised condition. Lower effort can indicate easier use, less engagement, or a different interpretation of the activity. The questionnaire alone does not determine which explanation applies. Similarly, the small difference in germane load does not show that the quality of learning was unchanged.

The separate three-item germane score is retained as exploratory in the workbook. It is not substituted for the two-item main score in Table 5.4. Legacy IPQ outcomes are not combined with SUS presence because the instruments and scoring issues differ.

5.6 Recall Time

For the 14 timed immediate-recall pairs, mean recall duration is 218.35 seconds in the generic condition and 259.69 seconds in the personalised condition. The mean difference is 41.34 seconds, with a 95% interval from -3.43 to 86.10 seconds and an unadjusted $p = .067$.

This does not show that the personalised palace is less efficient. Participants may spend longer because they recover more words or continue searching for additional items. Conversely, a long duration could reflect difficulty retrieving the route. The five-minute limit in the revised protocol also needs to be checked against the timing conventions and individual records. No delayed-duration comparison is reported because the workbook contains no complete timed pairs at those follow-ups.

5.7 Discussion

5.7.1 What the Current Comparison Can Say

The numerical pattern is compatible with a possible advantage for the personalised workflow, but it does not establish one. The immediate mean difference is positive, yet the participant-level directions are mixed and the confidence interval crosses zero. At later timepoints, differences remain uncertain and participant coverage decreases.

The findings should not be described as contradicting studies of VR immersion or active spatial binding. Those studies changed different aspects of the task [3, 2]. A system may benefit from active placement in both conditions while showing a smaller additional difference between a generic and a familiar palace. The present study is concerned with that additional difference.

5.7.2 Familiarity, Construction, and Exposure

The personalised condition combines several experiences. Participants bring prior knowledge of their home, actively edit the virtual environment, and spend additional time working with it. Any eventual difference between conditions could reflect one or more of those factors.

The familiar-layout rationale is supported by research on object and spatial memory [4], but this does not identify familiarity as the cause of the present numerical pattern. A stronger follow-up design could compare a generic palace, a familiar palace prepared for the participant, and an unfamiliar palace that the participant edits. Exposure time would also need to be measured or controlled.

5.7.3 Mixed Materials and Ceiling Effects

Pilot development aimed to avoid a task that participants could complete perfectly in both conditions. Nevertheless, the current snapshot contains both earlier and harder lists. Pooling them may combine different levels of task difficulty and different opportunities for improvement.

Normalised accuracy addresses the arithmetic difference between 20 and 25 targets, but does not remove differences in the words or their mnemonic images. The final analysis should therefore report results for the intended main-study materials and examine sensitivity to the inclusion of earlier sessions. Any version-based subgroup analysis will be exploratory unless its role was specified before examining outcomes.

5.7.4 Missingness and Order

The decrease from 18 immediate pairs to 16 at 24 hours and 14 at one week changes the sample being described. The missing participants may differ in motivation, availability, or memory performance. This draft does not assume that follow-up data are missing at random.

Order is another unresolved issue. The workbook’s session-order diagnostic contains no complete pairs, despite planned counterbalancing. That is not evidence that there was no order effect; it means the necessary diagnostic is currently unavailable. Recovering actual order from session records is required before practice and fatigue can be assessed.

5.7.5 Practical Value and Remaining Evidence

A personalised palace may be useful even if its mean recall advantage is modest, for example if users prefer to return to it over time. Conversely, a short-term recall gain may not justify a long creation stage. The current dataset cannot settle that trade-off without verified preparation-time, repeated-use, and preference evidence.

The system contribution is therefore distinct from the efficacy claim. The documented workflow shows how a familiar environment can be prepared, edited, and used for cue placement. Whether that workflow improves recall enough to justify its costs remains an empirical question. Final conclusions should follow the reconciled records, including the possibility of an inconclusive or small difference.

Chapter 6

Conclusion

6.1 Achievements

This thesis presents a VR memory palace workflow centred on spatial personalisation. The user prepares a layout based on their home, reviews and edits the walls, adds furniture, and uses the resulting environment to place visual mnemonic cues. The development records describe the practical work needed to support this process, including furniture editing, saving and loading, content import, and improvements to menu and wall interactions.

The evaluation compares this personalised workflow with a generic palace within the same participants. It considers recall quantity and order alongside cognitive load, enjoyment, competence, effort, and presence. Pilot development helped identify overly easy materials and an excessive questionnaire burden, leading to revised word lists and a shorter presence measure.

The current numerical snapshot contains a positive mean personalised-minus-generic difference in immediate recall accuracy of 6.28 percentage points across 18 complete pairs. However, the confidence interval crosses zero, the direction of the difference varies across participants, and the workbook remains blocked for interpretation. This draft therefore makes no final claim that personalisation improves recall.

6.2 Limitations

The principal limitation is the unresolved state of the analysis data. Duplicate questionnaire records and unreviewed responses must be reconciled before the results can support a final conclusion. The snapshot also combines materials versions, while the distinction between pilot and main-study sessions requires confirmation.

The experimental contrast combines familiarity with active authoring and extra exposure. It cannot isolate which of those components produces a difference. Follow-up observations are incomplete, actual condition order has not been recovered for the workbook’s diagnostic, and nominal follow-up labels do not establish precise retention intervals.

The system description is grounded in the available development logs rather than an audit of the final Unity build. Hardware, software versions, screenshots, recruitment details, and the exact administered questionnaire forms remain necessary additions. These gaps are recorded in Appendix A so that they can be resolved without turning assumptions into facts.

6.3 Future Work

The immediate priority is to complete the existing study record. This includes reviewing response decisions, resolving duplicate records, verifying participant and condition assignments, and identifying the final analysis cohort. The analysis should then be rerun with explicit reporting of missingness, materials versions, and condition order.

A subsequent experiment could separate familiarity from construction. Krokos et al. [3] explicitly proposed studying user-built palaces and comparing highly personal with shared palaces; the present study cannot resolve both questions because its home condition also involves authoring and extra exposure. Crossing familiar versus unfamiliar layouts with self-edited versus prepared layouts, while matching exposure time, would estimate those effects separately. Yang et al.’s participant feedback [13] and Perera et al.’s proposed preference-based templates [?] also motivate measuring whether users prefer the palace they create. Repeated-use sessions could then examine whether the initial preparation effort becomes worthwhile over several learning tasks and whether reusing loci causes interference.

The broader proposal also includes learning-content personalisation and visual and auditory mnemonic generation. These remain useful directions, but should be evaluated separately from the spatial comparison. Moll and Sykes [1] allowed a limited choice of images and suggested testing more personally chosen cues. A content study could hold the environment constant while varying cue choice or how the material is segmented and represented. An auditory study could compare visual cues with audio or combined cues while preserving the same target material and exposure time.

Finally, educational use should be evaluated using the outcomes it is intended to improve. Remembering a word list provides a controlled first test, but learning concepts also involves understanding and applying them. A later study should therefore include comprehension and transfer tasks rather than infer educational effectiveness from recall alone.

6.4 Closing Remarks

The project turns the idea of a familiar memory palace into an editable VR activity. Its central question is whether that personal connection to the space improves the experience and the memory task enough to justify the work of creating it. The present draft establishes the system account, evaluation structure, and limits of the available evidence. A final answer to the research question depends on completing and reconciling the study data.

Appendix

Appendix A

Draft Completion and Source Record

A.1 Purpose of This Draft

This first draft retains the broader proposal title while reporting the current spatial-personalisation study. Literature claims are drawn from the supplied papers and checked primary-source records. Project claims are drawn from the local protocol, proposal, development logs, and meeting notes. Numerical results refer to the workbook snapshot retrieved on 22 September 2026.

The supplied file named *my template paper.pdf* contains the Moll and Sykes article also supplied under its descriptive title. It is treated as one reference, not two independent studies. The research-notes document was used to locate additional literature; its informal summaries were not treated as a substitute for the original papers. The separate linked *Current Logs* Notion site could not be retrieved, so no facts are attributed to its unread contents.

A.2 Items to Confirm Before Submission

Table A.1: Information and checks required to complete the thesis.

Area	Required completion
Title and supervision	Confirm whether the approved broad title should remain for a thesis focused on the spatial study. The proposal names Dr. Yomna Mahmoud. Tobias Wagner’s involvement is documented in meeting logs, but his formal title-page designation needs confirmation.

Area	Required completion
Participants	Add recruitment, total recruited, demographics, eligibility, compensation, withdrawals, and the pilot/main-study boundary.
Ethics and consent	Add the actual institutional review status, consent process, data handling, and discomfort/withdrawal procedure. Do not substitute generic statements for the procedure used.
Apparatus	Record the headset and controller models, study location, application build, Unity version, locomotion settings, and relevant display/performance configuration.
Materials	Archive the exact ordered word lists and image options used by each session. Establish which versions belong in the final primary comparison.
Preparation	Confirm layout conversion format and workflow for the study build, creation time, furniture guidance, and any assistance given by the researcher.
Procedure	Confirm encoding and recall durations per version, whether recall was spoken or entered by participants, questionnaire timing, and whether the palace was visible during recall.
Counterbalancing	Reconstruct actual condition order and list assignment separately; report counts for each realised schedule.
Follow-ups	Verify the actual delay windows, delivery method, response instructions, rehearsal rules, and incomplete follow-up reasons.
Questionnaires	Verify the exact forms, cognitive-load source, endpoint coding, reverse scoring, creation-load measure, and legacy IPQ treatment.
Data quality	Resolve six reported blocking errors and 130 unreviewed, uncounted responses, then rerun the workbook and inspect its approval status.
Implementation evidence	Add screenshots from the actual study build and verify implementation details against the Unity source. The functional diagrams in this draft are not application screenshots.
Final reporting	Update all tables, abstract values, discussion, and conclusions from the same approved dataset. Review acknowledgments and the unsigned declaration.

A.3 Recorded Protocol Changes

Table A.2: Selected milestones used to reconstruct the evolving study. Dates identify log entries rather than independently verified deployment dates.

Record	Evidence
Early protocol	30 words, 15-minute encoding, IPQ presence, and a 24-hour follow-up [30].
17 July 2026	20-word lists with five easy, ten moderate, and five hard words; revised image-search ideas and wall editing [19].
27 July 2026	Customisation before either condition; break and cognitive-load assessment after creation [23].
10–12 August 2026	Recommendation and logged change to SUS presence; switching lists between conditions [21, 22].
13 August 2026	25-word balanced lists with ten easy, ten medium, and five hard words [20].
Local revised notes	25 low-memorability words, ten-minute encoding, questionnaires before five-minute verbal recall. Exact start of use remains to be verified.
22 September 2026	Analysis snapshot includes Condition List, Memory List, and Hard Memory List records; data-quality approval remains blocked [8].

A.4 Numerical Reproducibility

The file `data/results_snapshot_2026-09-22.json` records the aggregate key results and the recall fields needed to check their arithmetic. Participant identifiers in this local file are pseudonymous. It contains neither participant names nor raw response text.

Running `verify_snapshot.py` with Python checks complete-pair counts, trial accuracy denominators, condition means, paired differences, paired t statistics, and d_z for the three recall timepoints. The script uses only the Python standard library. It does not re-adjudicate responses, validate exclusions, or independently recalculate the workbook’s p values and confidence intervals.

The spreadsheet tabs used for this draft are *V3 Start Here*, *V3 Key Results*, *V3 Data Quality*, *V3 Recall Trials*, *V3 Session Order*, and *V3 Data Dictionary*. The supplied deep link’s numeric tab identifier was absent from the current metadata; the named tabs were resolved from the workbook rather than guessed.

Appendix B

Lists

List of Abbreviations

AI	Artificial Intelligence
AR	Augmented Reality
CI	Confidence Interval
CSV	Comma-Separated Values
HMD	Head-Mounted Display
IMI	Intrinsic Motivation Inventory
IPQ	Igroup Presence Questionnaire
JSON	JavaScript Object Notation
LCS	Longest Common Subsequence
MoL	Method of Loci
SD	Standard Deviation
SUS	Slater–Usch–Steed presence questionnaire
VR	Virtual Reality

List of Figures

3.1	Revised procedure reconstructed from the local protocol and supervisory notes. Earlier sessions used different materials; actual follow-up timing remains to be verified.	16
4.1	Functional overview of the documented workflow. Analysis includes records collected outside the VR application.	22
5.1	Paired recall differences in the frozen snapshot. Every interval crosses zero. The workbook remains blocked for interpretation pending data review. .	29

List of Tables

2.1	Selected related work and its relationship to the current study.	11
3.1	Intended counterbalancing structure, not verified allocation counts. G denotes generic and P denotes personalised.	17
3.2	Questionnaire scoring definitions in the current analysis workbook.	18
5.1	Complete pairs available in the provisional workbook snapshot.	28
5.2	Provisional recall accuracy as a percentage of the trial's target list. SD denotes sample standard deviation.	28
5.3	Provisional paired recall comparisons. Differences and confidence intervals are in percentage points, personalised minus generic. All p values shown are unadjusted.	29
5.4	Provisional questionnaire results. G denotes generic and P personalised; $\Delta = P - G$. Confidence intervals and p values concern paired differences and are unadjusted.	30
A.1	Information and checks required to complete the thesis.	39
A.2	Selected milestones used to reconstruct the evolving study. Dates identify log entries rather than independently verified deployment dates.	41

References

- [1] Brigham Moll and Ed Sykes. Optimized virtual reality-based Method of Loci memorization techniques through increased immersion and effective memory palace designs: a feasibility study. *Virtual Reality*, 27:941–966, 2023. <https://doi.org/10.1007/s10055-022-00700-z>.
- [2] Nicco Reggente, Joey K. Y. Essoe, Hera Younji Baek, and Jesse Rissman. The Method of Loci in virtual reality: Explicit binding of objects to spatial contexts enhances subsequent memory recall. *Journal of Cognitive Enhancement*, 4:12–30, 2020. <https://doi.org/10.1007/s41465-019-00141-8>.
- [3] Eric Krokos, Catherine Plaisant, and Amitabh Varshney. Virtual memory palaces: immersion aids recall. *Virtual Reality*, 23:1–15, 2019. <https://doi.org/10.1007/s10055-018-0346-3>.
- [4] Niamh A. Merriman, Jan Ondřej, Eugenie Roudaia, Carol O’Sullivan, and Fiona N. Newell. Familiar environments enhance object and spatial memory in both younger and older adults. *Experimental Brain Research*, 234(6):1555–1574, 2016. <https://doi.org/10.1007/s00221-016-4557-0>.
- [5] Mark Sroor. Research proposal: Personalized multimodal mnemonics in immersive VR memory palaces. <https://www.notion.so/37954328636780af82f4f47accf5e3e0>, 2026. Proposal, not a report of completed experiments. Accessed 22 September 2026.
- [6] Mark Sroor. Supervisory meeting record, 20 january 2026. <https://www.notion.so/16f5432863678398ad9301167f5d3794>, 2026. Scope and separate evaluation of environment and content personalisation. Accessed 22 September 2026.
- [7] Martin Usoh, Ernest Catena, Sima Arman, and Mel Slater. Using presence questionnaires in reality. *Presence: Teleoperators and Virtual Environments*, 9(5):497–503, 2000.
- [8] Mark Sroor. V3 analysis: research workbook. https://docs.google.com/spreadsheets/d/1_Y-vJ2na7T52dyh9PWdt_ahwo-QuYVgeRKjSNdpm09w/edit, 2026. Unpublished study data. Snapshot retrieved 22 September 2026; data-quality interpretation gate blocked.

- [9] Center for Self-Determination Theory. Intrinsic motivation inventory (IMI). <https://selfdeterminationtheory.org/intrinsic-motivation-inventory/>. Instrument documentation. Accessed 22 September 2026.
- [10] Eric L. G. Legge, Christopher R. Madan, Enoch T. Ng, and Jeremy B. Caplan. Building a memory palace in minutes: Equivalent memory performance using virtual versus conventional environments with the Method of Loci. *Acta Psychologica*, 141(3):380–390, 2012. <https://doi.org/10.1016/j.actpsy.2012.09.002>.
- [11] Nicko R. Caluya, Xiaoyang Tian, and Damon M. Chandler. Comparison of AR and VR memory palace quality in second-language vocabulary acquisition. *Electronic Imaging*, 35(10):220–1–220–5, 2023. <https://doi.org/10.2352/EI.2023.35.10.HVEI-220>.
- [12] Samuel Sandberg Bröms. Objects vs images for Method of Loci in mixed reality: Comparing the efficacy of using 3D objects against 2D images in the Method of Loci. Degree project, KTH Royal Institute of Technology, 2024. Supplied thesis, dated 17 December 2024.
- [13] Fumeng Yang, Jing Qian, Johannes Novotny, David Badre, Cullen D. Jackson, and David H. Laidlaw. A virtual reality memory palace variant aids knowledge retrieval from scholarly articles. *IEEE Transactions on Visualization and Computer Graphics*, 27(12):4359–4373, 2021. <https://doi.org/10.1109/TVCG.2020.3009003>.
- [14] Samarth K. Reddy. From method of loci to digital mnemonics: AI, AR, and the future of memory. Master of design thesis, OCAD University, 2025. <https://openresearch.ocadu.ca/id/eprint/4639/>.
- [15] Lei Zhang, Jin Pan, Jacob Gettig, Steve Oney, and Anhong Guo. VRCopilot: Authoring 3D layouts with generative AI models in VR. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. ACM, 2024. <https://doi.org/10.1145/3654777.3676451>.
- [16] Christopher R. Madan, Mackenzie G. Glaholt, and Jeremy B. Caplan. The influence of item properties on association-memory. *Journal of Memory and Language*, 63(1):46–63, 2010. <https://doi.org/10.1016/j.jml.2010.03.001>.
- [17] Christopher R. Madan. Exploring word memorability: How well do different word properties explain item free-recall probability? *Psychonomic Bulletin & Review*, 28:583–595, 2021. <https://doi.org/10.3758/s13423-020-01820-w>.
- [18] Mark Sroor. Masters logs: development and meeting records. <https://www.notion.so/27254328636780c8b6dcc8d494b53f0b>, 2026. Unpublished project records. Accessed 22 September 2026.
- [19] Mark Sroor. Development log, 17 july 2026: list materials and wall-maker improvements. <https://www.notion.so/3a05432863678011a173ce4126c23849>, 2026. Unpublished project record. Accessed 22 September 2026.

- [20] Mark Sroor. Development log, 13 august 2026: 25-word lists. <https://www.notion.so/3bb54328636780c6a37efbd1140de0d1>, 2026. Unpublished project record. Accessed 22 September 2026.
- [21] Mark Sroor. Supervisory meeting record, 10 august 2026. <https://www.notion.so/3b8543286367804db623f3306b1a9d6d>, 2026. Presence questionnaire and list assignment. Accessed 22 September 2026.
- [22] Mark Sroor. Development log, 12 august 2026: alternating lists and SUS questionnaire. <https://www.notion.so/3b954328636780f7a1bde5a34d0cb73c>, 2026. Change recorded in entry title. Accessed 22 September 2026.
- [23] Mark Sroor. Supervisory meeting record, 27 july 2026. <https://www.notion.so/3aa543286367806493f9d4d186283768>, 2026. Creation stage and cognitive-load timing. Accessed 22 September 2026.
- [24] Mark Sroor. Development log, 18 december 2025: saving, loading, and furniture editing. <https://www.notion.so/eae5432863678263a41c017e045afca0>, 2025. Unpublished project record. Accessed 22 September 2026.
- [25] Mark Sroor. Development log, 5 october 2025: mapping a board to walls. <https://www.notion.so/7d554328636783ec8b4e01aca0035117>, 2025. Prototype recorded in entry title and images. Accessed 22 September 2026.
- [26] Mark Sroor. Development log, 28 november 2025: integrated furniture and room placement. <https://www.notion.so/97554328636783cdb6b0012821c28bbc>, 2025. Unpublished project record. Accessed 22 September 2026.
- [27] Mark Sroor. Development log, 13 april 2026: importing loci from JSON. <https://www.notion.so/d8554328636783eeb263013281fbc085>, 2026. Includes content-entry example. Accessed 22 September 2026.
- [28] Mark Sroor. Development log, 13 march 2026: placing loci. <https://www.notion.so/9a954328636783be85eb01de964295c8>, 2026. Prototype recorded in entry title and image. Accessed 22 September 2026.
- [29] Mark Sroor. Development log, 15 may 2026: revising locus placement. <https://www.notion.so/e715432863678348afad013ef06c07f4>, 2026. Change recorded in entry title. Accessed 22 September 2026.
- [30] Mark Sroor. Experiment protocol. <https://www.notion.so/38a543286367805b969cc742fb791357>, 2026. Historical protocol; differs from the revised local procedure. Accessed 22 September 2026.